

Exercise 3

The Bohr magneton and the nuclear magneton are

$$\mu_B = 5.79 \times 10^{-5} \text{ eV/T}, \quad \mu_N = 3.15 \times 10^{-8} \text{ eV/T}.$$

- Calculate $\mu_B B$ and $\mu_N B$ for $B = 1 \text{ T}$.
- Find the ratio $\frac{\mu_B}{\mu_N}$.
- Compare both energies with $k_B T$ at $T = 1 \text{ K}$.

Explain why electron spins are much easier to polarize than nuclear spins.

Exercise 4

Consider N non-interacting spin- $\frac{1}{2}$ magnetic moments μ in an external magnetic field B . The two energy levels are

$$E_{\uparrow} = -\mu B, \quad E_{\downarrow} = +\mu B.$$

- Show that the single-spin partition function is

$$Z_1 = e^{\beta\mu B} + e^{-\beta\mu B} = 2 \cosh(\beta\mu B), \quad \beta = \frac{1}{k_B T}$$

- Hence find the free energy of the whole system:

$$F = -Nk_B T \ln[2 \cosh(\beta\mu B)].$$

- Using

$$S = - \left(\frac{\partial F}{\partial T} \right)_B,$$